HiRes Fieldbook

APPN Aerial SOP — Locked revision 1.03

APPN – Hi-Res Fieldbook

For a high-level description of the Hi-Res sensor package and how it sits alongside the other APPN platforms, see the Platforms Overview.

This fieldbook provides a standardised operational guide for APPN Hi-Res UAV deployments using the Phase One iXM-GS120 120 MP RGB camera mounted on a DJI M350 RTK. It is intended for trained APPN staff conducting high-resolution RGB data acquisition for crop emergence, canopy architecture, and structural phenotyping applications. The protocol supports safe flight operations, consistent sensor configuration, and high-quality, repeatable data capture across APPN field nodes.

Important

This protocol must be followed for all standard APPN Hi-Res UAV flights. Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Season Working Plan — Team & Outstanding Items

Note

This section captures the cross-node working arrangements and outstanding decisions agreed for the current season. It is project / coordination context rather than field procedure, and is mirrored in TODO_FUTURE.md for tracking.

Team responsibilities (proposed — to be confirmed by each node)

Person / Team	Proposed contribution
UQ	Lead the photogrammetry pipeline build; contribute time and expertise.
Author (technical)	Contribute to the technical side of photogrammetry processing and Metashape settings.
James	Support pipeline development.
Dillon	Technical review (proposed).
Richard / Arden (USyd)	Support and advise (proposed).
CSU team	Provide technical input (proposed).
Bipul	Integration of tooling for image conversions, etc. (proposed).
Warin & Bipul	Lead non-photogrammetry pipeline; expand team as needed.

Critical outstanding items

- ☒ **APPN plot-extraction method** — agreed; see Plot Delineation.
- ☐ **Working team confirmation** — node leads to confirm named contributors for the photogrammetry pipeline.
- ☐ **PhaseOne SDK capability** — confirm scope of Image SDK functionality and resourcing for GUI / code expertise.
- ☐ **Boresight calibration** — decide whether to pursue as a means of improving image geo-location.

Document Structure

1. Equipment Checklist — what to bring to the field.
2. First Use and Preseason Preparation — pre-deployment commissioning of the camera, gimbal, and aircraft.
3. Preflight Planning — area of interest, KML creation, and DJI Pilot 2 mission setup.
4. APPN Hi-Res Mission Standard Types — Table 1 standard mission bundles.
5. Camera Capture Settings — aperture, shutter, ISO, and focus, including setting exposure with the live histogram.
6. Onsite Preflight Operations — weather and airspace checks, GCP layout, payload mount, and aircraft setup.
7. Flight Operations — takeoff, autonomous mission, and landing.
8. In-Flight Data Confirmation — image count, exposure, and storage checks.
9. Post-Flight — pack-down of aircraft, payload, and ground reference kit.
10. Offload Data — downloading from the camera and storing under the APPN folder structure.
11. Sensor Configuration Reference — Table 2 consolidated mission, camera, and environmental settings.
12. GSD vs Altitude Lookup — Table 3 ground sampling distance reference.
13. Non-Routine, First-Time, and Commissioning Procedures — damping plate, iX Capture, gimbal balancing, CoG calibration, and custom camera setup.

Equipment Checklist

> **Note**
>
> Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

- ☐ **Aircraft**
- ☐ DJI M350 (with P3 gimbal damping plate attached)
- ☐ Aircraft batteries and spares
- ☐ Landing gear
- ☐ Landing pad
- ☐ Spare parts
- ☐ Tools
- ☐ Logbook
- ☐ **Radio Control Transmitter / Ground Control Station**
- ☐ **Hi-Res sensor payload**
- ☐ Phase One iXM-GS120 camera body and P3 gimbal
- ☐ 80 mm lens
- ☐ Camera storage media (CFexpress) and card reader
- ☐ **Ground reference kit**
- ☐ DJI D-RTK 2 base station, tripod and spare batteries
- ☐ Ground control points
- ☐ GNSS system (Emlid / Trimble)
- ☐ Wooden base station peg with marked centre-point (optional — for when returning to a location for surveys through time, or for linking to real-world GNSS data)
- ☐ **Accessories**
- ☐ Safety gear (signage and high-vis vests)
- ☐ Aeronautical radio
- ☐ Field laptop and spare batteries
- ☐ External storage media
- ☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
- ☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
- ☐ External power brick (for charging UAV RC)
- ☐ Handheld anemometer (such as a Kestrel)

☐ Professional lens cleaning wipes

First Use and Preseason Preparation

Important

Performance of the camera relies on correct installation of the Phase One camera within the gimbal, a perfectly balanced gimbal, and the drone performs best when the camera weight and centre of gravity has been factored into the DJI flight control software.

Please refer to Non-Routine, First-Time, and Commissioning Procedures at the end of this document for information on installing the damper plate, gimbal installation and setup, gimbal balancing, and drone centre of gravity balancing.

Note

Once you have set the gimbal balance, note the settings and offsets and take photos of the setting locations to ensure you have reference settings for a stable gimbal. If the camera and gimbal go out of balance, immediately inspect the balance settings and reset them to the known default balanced values for that camera.

Preflight Planning

Important

Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance. Further planning documentation for the Phase One payload on the M350 can be found in the Phase One iX Cameras on DJI M300 with P3 Gimbal Installation Guide. Note that before first flight with this sensor, several commissioning steps must be completed — see Non-Routine, First-Time, and Commissioning Procedures.

1. Using a GPS survey system (Emlid, Trimble…) or GIS software, create a polygon of the area of interest. Make sure your polygon includes the areas where you will place your GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Save the polygon as a KML file using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
3. Load the KML file onto a USB stick or microSD card for transfer to the M350 controller.
4. Power on the controller.
5. Open DJI Pilot 2 and select **Flight Route → + Import Route (KMZ/KML)**. Alternatively, you can create an Area Route on the controller (e.g. by walking the field). *Note that you do not need the UAV powered on for this — just the controller.*
6. Confirm that the Phase One iXM-GS120 payload is correctly detected in DJI Pilot 2 and selected as the active payload for the mapping mission. Ensure the correct aircraft model (M350 / M300 RTK) is selected, and configure a custom camera profile using the appropriate Phase One P3 (80 mm) parameters.



7. Set the flight altitude mode to **Relative to Take-off Point**.
8. Set image overlap and triggering parameters in the **Advanced Settings** of the mapping mission:

- Front (forward) overlap: **80%**
- Side (lateral) overlap: **70%**
- Photo mode: **Distance Interval Shot**

> **Important**
>
> 80/70 overlap always produces a reasonable motion blur <0.5 GSD.

9. Set the course angle to align with the experimental layout:

- Use **row-aligned** orientation for plot-based trials where crop rows are clearly defined.
- Use **North–South** orientation for broad-scale paddock surveys.

10. Select the flight altitude and associated ground sampling distance (GSD) and flight speed according to the mission standard type settings in Table 1 — APPN Hi-Res Mission Standard Types. These bundles are optimised to balance spatial resolution, coverage, and motion-blur risk.
11. Disable **Elevation Optimisation**.
12. For flight settings including height and pixel size (including ground sampling distance (GSD)), and flight speed (to avoid motion blur), refer to Table 1 — APPN Hi-Res Mission Standard Types, and use DJI Pilot 2 app. If a custom setup is needed, refer to the Phase One flight calculator to optimise flight speed whilst minimising blur.
13. Save the mission using the convention `YYYYMMDD_XXXX` (as above).

APPN Hi-Res Mission Standard Types

Table 1 — APPN Hi-Res Mission Standard Type settings.

Pre-tuned parameter bundles for the four APPN Hi-Res Mission Standard Types. Flight parameters must match the application and must not deviate from these settings.

Standard Mission Type	Scenario	Altitude	Speed	Shutter	Aperture	Overlap (Front-lap/Sidelap)
Type 1	Plant counting / small structures (flowers, heads, early lesions, emergence counting)	12-20 m	0.9–1.6 m/s	≥ 1/4000	f/8	80/70
Type 2	Plot phenotyping (canopy traits, vigour)	30–50 m	2.4–4.1 m/s	≥ 1/4000	f/8	80/70
Type 3	Canopy coverage / plot averages	50–70 m	4.1–5.7 m/s	≥ 1/4000	f/8	80/70
Type 4	Large-area flat terrain ortho (efficiency mode)	80–120 m	6.5–9.8 m/s	≥ 1/4000	f/8	80/70

Note

80/70 overlap is an acceptable setting for orthomosiac workflows and for larger field sites. It may not be enough overlap for other approaches. See the HiRes Processing Pipeline for the processing methods these settings feed into.

Camera Capture Settings

Press the camera image icon in DJI Pilot 2 and configure the camera appropriate to your APPN Hi-Res Mission Standard Type. For full per-parameter guidance, see Sensor Configuration Reference (Table 2).

1. **Aperture** — set to a fixed value of **f/8** across all operating scenarios. The fixed aperture provides a moderate depth of field while keeping shutter speed fast enough to prevent motion blur in constant light conditions.
2. **Shutter speed** — set a **minimum shutter speed of 1/4000 s** across all operating scenarios per Table 1. DJI Pilot 2 mission planning will always set a conservative shutter speed that does not allow image motion blur.
3. **Focus** — default mode is to set the focus mode to **Auto Focus (AF)**. This uses the Laser Range Finder (LRF) mounted on the camera to set the focal distance. This is the optimal setting for most applications.

Note

Some crop types and applications, such as imaging plant heads at maturity in plots surrounded by bare soil where the focal distance to the target of interest may change when using AF mode, advanced focal settings may be required to ensure the target of interest is always in focus. Consider setting **Manual Focus (MF)** and "Focus Distance Slider" to set focus at the distance to the target where this is consistent (e.g. on flat ground or on terrain-following missions). You can also use advanced "Focus Limit Control" focus settings and "Gate" (or other) options with an appropriate minimum and maximum focus limits. It is not recommended to use "Focus Bracketing" solutions, as this will generate many additional images at the same location that will impact photogrammetry processing and also impact flying speed.

Also refer to the P3 Camera advanced focus settings for more information here.

4. **ISO range** — Min: **200** / Max: **3200**. **ISO is the only variable parameter** used to achieve the correct exposure and a balanced histogram. The minimum ISO available on the camera is **ISO 200**, and the maximum recommended ISO is **ISO 3200** to avoid introducing excessive image noise. As a general reference, under clear-sky conditions around solar noon the expected ISO value may be around **ISO 1250**, although this will vary with actual lighting conditions, cloud cover, crop reflectance, altitude, and mission timing. See Setting exposure with the live histogram below for the on-site procedure.
5. Display the **live histogram** and confirm there is no sustained clipping in the highlights or shadows. Do **not** use any **EV compensation (+/-)** — allow ISO to adjust automatically within the configured range.
6. Before take-off, **test-fire the camera** (from the DJI Pilot 2 app and M350 Controller) and confirm that the image is written successfully to the storage media.
7. Ensure any legacy data has been downloaded and backed up. Format the memory card from the DJI Pilot 2 app menus within the M350 Controller.
8. Before starting the mission, manually fly to the planned operating altitude with the camera pointing down at the crop / plants. Confirm that the live preview and histogram show correct exposure.

Note

For more information on flight settings and what they do, refer to Sensor Configuration Reference (Table 2).

Setting exposure with the live histogram

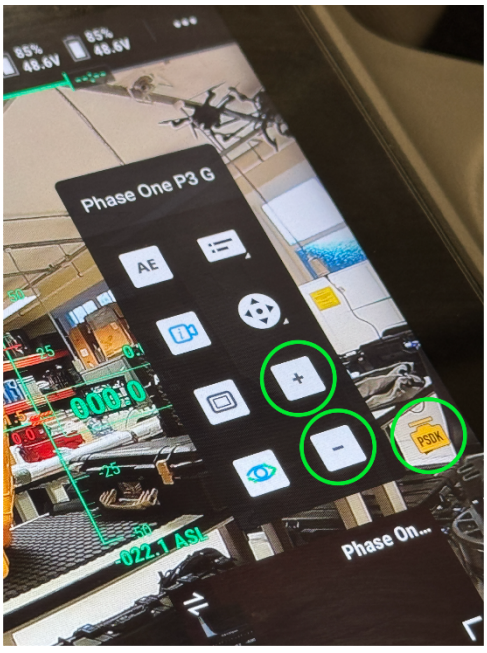
In constant light conditions, the most consistent image capture comes from setting a fast shutter speed (to prevent motion blur) and a fixed aperture (f/8) that gives a moderate depth of field, and using **ISO as the only variable** to balance exposure.

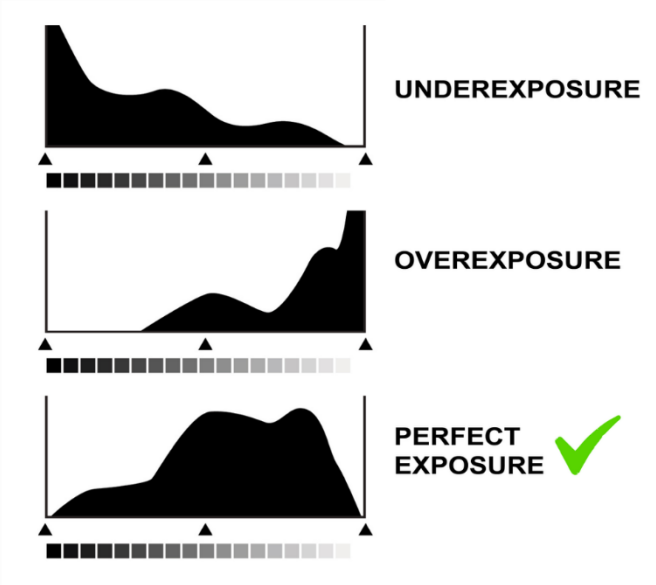
To check and set the correct exposure before starting the mission:

1. Set a minimum shutter speed of **1/4000 s** and an aperture of **f/8** in the DJI Pilot 2 camera settings.
2. Fly the aircraft to the area of interest at the proposed mission altitude.
3. Point the camera at nadir (straight down) over the target crop or survey area.
4. Review the live histogram (bottom-left of the controller screen).
5. Adjust the ISO using the exposure control **(+ / -)** buttons on the right-hand side of the controller screen.
6. Continue adjusting until the histogram shows a suitable exposure without clipping at either end.



A correctly exposed histogram may appear as a single smooth peak, or as a series of peaks and valleys. Both are normal, depending on the scene. The key point is to monitor the left and right edges of the histogram and avoid clipping — particularly on the right side, where overexposed highlights result in lost image detail.





The + and − buttons are used to adjust exposure values. In this protocol, those buttons are used to increase or decrease the **ISO** value to achieve the correct histogram exposure (shutter speed and aperture remain fixed).

Onsite Preflight Operations

1. Conduct airspace and weather checks.

- Check NOTAM (find in ERSA).
- Ideally, no cloud cover. Avoid rapidly changing cloud conditions where possible.
- Maximum wind speed for quality capture is **6 m/s (22 km/h)**. Any wind speed over **5 m/s (18 km/h)** should be recorded.
- Do not operate in conditions below 0 °C or above 40 °C.

2. Turn on the aeronautical radio and set to local CTAF (find in ERSA).
3. Set up the DJI D-RTK 2 base station on a tripod and power on at least 15 minutes prior to flight. Ensure it is set up in **operation mode 5** (five LED blinks for communication with the M350).

For any missions returning to the same location or requiring alignment to non-DJI data, the DJI D-RTK 2 base station must be set up over a known point that is surveyed against a GNSS base station (e.g. using AUSCORS or AUSPOS), or set to the location used for a previous survey. In the M350 controller RTK settings, manually edit the latitude and longitude of the base station to the known surveyed (or past) location. The aircraft should be switched off during this step.

4. Deploy GCPs in the field, ensuring good coverage of the area of interest. A **minimum of 6 Aeropoints (or other GCPs)** should be deployed per site. If using Propeller Aeropoints, ensure they are switched on and logging. If using standard GCPs, log GNSS coordinates using a Trimble or Emlid GNSS system.
5. Set up the landing pad and UAV in a safe RTH location.

- In dusty environments an additional tarp should be used under the landing pad.

6. Attach the Hi-Res payload to the UAV:

- Check that the lens and sensors are clean. If not, use professional lens cleaning wipes to clean them.
- Check the damping plate to ensure it is mounted securely to the UAV and that the USB-C cable is connected.
- Remove the protective metal cap from the quick-release receptacle.
- Insert the gimbal into the quick-release receptacle on the damping plate, aligning the unlocked symbol on the bayonet opposite the white mark on the receptacle.
- Rotate the non-slip ring to the right until the locked symbol is opposite the white mark on the receptacle and the bayonet clicks and locks. *Note: this mount is tool-less, but correct alignment is critical.*

- **Remove the lens cap.**
- Ensure the CFexpress card is installed and that there is adequate space to store mission data.

7. Power on the radio controller; check battery status.
8. Launch DJI Pilot 2.
9. Load and review the flight plan, checking operational height and double-checking that the area of interest, GCPs, and reflectance panels are all within the capture polygon.
10. Power on the aircraft; confirm connection to RC/GCS and battery status.
11. On the controller, set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings. Check that the mission is available on the controller.
12. If you are using RTK, check that you are receiving GNSS corrections. There should be more than 8 satellites for good correction. Check the RTK resolution in the M350 controller settings — the solution should be **FIX** with horizontal and vertical error less than **0.03 m (3 cm)**.

Flight Operations

1. Ensure the aircraft is in **Position** flight mode.
2. Clear people / objects away from the UAV.
3. Notify crew / observers that takeoff is beginning.
4. Begin manual takeoff, fly quickly to ~12 m AGL, and check stick controls work.
5. Enable the autonomous mission prepared earlier.
6. Monitor UAV battery voltage / percentage in flight.
7. After mission, switch back to **Position** mode to regain manual control.
8. Land and disarm the UAV.
9. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks.

In-Flight Data Confirmation

1. On the RC Plus controller, check image count and remaining storage on the card (to ensure the card did not fill up during the mission). Review the images as the mission proceeds. Ensure that the image count is increasing as images are written to the card. Listen for the "click" noise as images are captured. Review the histogram settings and the photos to ensure that images have appropriate illumination.

Post-Flight

1. Power off the drone, then the controller.
2. Inspect the drone for any damage and report for maintenance if necessary. Feel both batteries for excess or differential temperature. Feel the gimbal motors for excess or differential temperature. Inspect propellers, airframe, and landing gear.
3. Remove aircraft batteries.
4. Remove the payload and place it in its case.
5. Pack up the drone and all gear from site, double-checking against the checklist at the start of this document to ensure everything is accounted for.

Offload Data

1. Remove the CFexpress card from the camera and offload data to a computer using a card reader.
2. After offload and backup, format the card for the next mission in the DJI Controller.

Sensor Configuration Reference

Table 2 — Consolidated mission, airframe, camera, and workflow settings. All values assume the **80 mm AF lens** on the **iXM-GS120** body; substituting any other lens or camera body invalidates the GSD column in Table 3 and requires re-calculation.

Mission type & route setup (DJI Pilot 2 — Mapping)

Parameter	Recommended setting	Operational note
Mission Type	Area Route — Ortho Collection	Use the dedicated Ortho Collection template. Do not use Oblique Collection for orthomosaic output.
Ortho GSD (cm/pixel)	Leave default — auto-populates from Route Altitude	Altitude drives GSD via the Phase One sensor geometry.
Altitude Mode	Relative to Takeoff Point (ALT)	Assumes takeoff elevation ≈ site elevation.
Route Altitude (m)	Select from Table 3 — tailor to mission objective	Counting plant organs / flowers / heads → lowest practical altitude (12–20 m). Canopy coverage / plot averages → 50–120 m.
Elevation Optimisation	DISABLED	Do not enable for ortho missions — can cause alignment issues in Metashape.
Speed (m/s)	As fast as mission allows (recommended 3–10 m/s max); cross-check with Table 1	Motion blur = shutter interval (s) × drone speed (m/s). **Rule: motion blur ≤ GSD.** See worked example below.
Frontal & Side Overlap	80 / 80 (default, high-quality ortho)	80/80 is the true-ortho setting — minimises lean, best for crop / plant-structure work.
Route Start Point	Furthest polygon corner from pilot	Mission works back toward the pilot, finishing closest to the takeoff point. Reduces return-transit risk if battery runs low.
Takeoff Speed (m/s)	10–15	Transit-only speed (takeoff point → first waypoint). Safe to run faster than capture speed because no images are triggered in transit.

Tip

Worked example — 50 m altitude, 4 m/s, 1/4000 s:

- Shutter interval: $1 \div 4000 =$ **0.00025 s**
- Drone speed: **4 m/s**
- Motion blur: $0.00025 \times 4 =$ **0.001 m = 1.0 mm**
- GSD at 50 m: **2.1 mm** = 0.0021 m
- Check: 1.0 mm ≤ 2.1 mm **Pass** (blur is ~½ pixel — ideal).

Phase One iXM-GS120 (80 mm AF lens) — camera capture settings

Parameter	Recommended setting	Operational note
Exposure Mode	Manual (fixed aperture + fixed shutter + auto ISO)	Lock shutter (1/4000 or faster — see Table 1), set aperture to f/8, let ISO adjust. Protects motion-blur ceiling first, radiometric quality second.
EV Compensation	0 baseline — adjust to +1 in low-light / emergence conditions	+1 EV pushes the histogram right for more tonal information. Verify with live histogram.
ISO	Base 200 (sensor native) — max 3200	iXM-GS120 native ISO range is 200–6400 (RGB). Noise becomes visibly detectable above ISO 3200.
Aperture	f/8 min (sweet spot) — f/11 max	Sharpness peaks ~f/8 across the frame; edge softness appears wide of f/5.6 and diffraction softens past f/11.
Shutter Speed	Set per Table 1 — minimum 1/4000, up to 1/16,000	Motion blur rule: shutter × speed ≤ GSD × 0.5 (half-pixel tolerance).
Focus Mode	Auto Focus (AF) using the on-camera Laser Range Finder (LRF) as the default. Switch to Manual Focus with the Focus Distance Slider, or use Focus Limit Control (Gate) for targets where the LRF reading may not track the object of interest (e.g. tall heads above bare soil).	At f/8 the depth of field is sufficient for most APPN missions when AF/LRF is used. Avoid Focus Bracketing — it inflates frame count and breaks consistent GSD assumptions for ortho processing. See the Phase One P3 Smart Focus Features guide.
Focus Bracketing	OFF	Bracketing multiplies frame count and breaks consistent GSD assumptions for ortho processing.
White Balance	Fixed to Daylight (5500 K)	Fixed WB ensures radiometric consistency for post-processing.

Parameter	Recommended setting	Operational note
File Format	IIQ (RAW, 14-bit)	Expect ~100 MB per IIQ frame.

Environmental capture window

Parameter	Recommended setting	Operational note
Cloud Cover (preferred)	Clear blue sky **or** uniform high-altitude cover (cirrus / cirrostratus)	Both yield consistent illumination across the mission. Uniform altostratus is acceptable if genuinely even.
Cloud Cover (avoid)	Low-level broken / patchy cumulus	Hard shadows move across the AOI during capture → per-frame radiometric inconsistency.
Sun Angle and Capture Window	Sun within $\pm 20^{\circ}$ of solar noon (~2 hours before or after noon)	Solar noon depends on time of year and latitude — check the NOAA solar calculator if unsure.
Wind	$\leq$ manufacturer limit for M350 RTK (12 m/s sustained); prefer $\leq$ 8 m/s for photogrammetric quality	Gust-induced position jitter degrades overlap consistency.

GCPs, boresight & calibration

Parameter	Recommended setting	Operational note
Ground Control Points	Minimum 6 per site (4 corners + 2 distributed across the AOI); surveyed with RTK/PPK receiver, $\leq$ 2 cm accuracy	Required for absolute accuracy validation. For repeat monitoring plots, install permanent GCP markers.
Cross-Runs over GCPs	INCLUDE — fly a perpendicular cross-strip over the GCP line after the main mission	Per Phase One ANZ guidance, cross-runs are valuable for boresight angle determination, especially after CoG calibration and with the gimbal aligned to nadir. Adds ~5 min per mission.
Gimbal Calibration	Verify nadir alignment before each flight day	Even small gimbal offsets propagate into exterior orientation errors across a full mission.

GSD vs Altitude Lookup

Table 3 — Ground Sampling Distance reference for the Phase One iXM-GS120 with RSM 80 mm AF lens. Derived from pixel pitch (3.45 μm) $\times$ altitude / focal length (80 mm); values match the Phase One published GSD table.

Flight Altitude (m AGL)	GSD (mm/px)	GSD (cm/px)	Typical Use Case
12	0.5	0.05	Ultra-high detail — flower / head counting, disease lesions
20	0.9	0.09	Plant-organ ID, tillering, early-stage phenotyping
30	1.3	0.13	Plot-level phenotyping, canopy architecture
50	2.1	0.21	Standard orthomosaic — canopy coverage, NDVI-analog
70	~3.0	0.30	Large-area plot-average traits
100	~4.3	0.43	Site-scale mapping, large trial blocks
120	5.2	0.52	Wide-area ortho — max altitude standard ops

Non-Routine, First-Time, and Commissioning Procedures

Damping plate installation

For full details, see the Phase One P3 Type-I Gimbal Integration Guide.

Also refer to the video: Mounting the P3 Payload | Phase One.



The P3 gimbal requires a custom damping plate on the front of the M350 UAV.

1. Power down the aircraft and remove batteries.
2. Place the aircraft in a protective cradle and invert.
3. Remove the USB-C connector screws.
4. Disconnect the USB-C connector.
5. Remove the DJI factory damping plate.
6. Install the Phase One P3 damping plate.
7. Apply Loctite 222 to all mounting screws.
8. Torque mounting screws to specification.
9. Reconnect the USB-C connector.
10. Torque the USB-C screws.
11. Return the aircraft upright.

iXM camera configuration via iX Capture

For full details, see the Phase One P3 Type-I Gimbal Integration Guide.

1. Insert storage media into the camera.
2. Connect the camera to a PC via USB-C.
3. Power the aircraft to supply camera power.
4. Open **iX Capture**.
5. Format storage media.
6. Set **LINK type = DJI M300 / M350**.
7. Restart the camera.
8. Confirm the camera reconnects successfully.

Gimbal Balancing

Balancing of the gimbal is critical to camera performance, including risks from blurred images and gimbal motor overheating. The gimbal must be balanced such that the camera remains in any neutral position it is placed in. Follow the steps below and the instructional video on gimbal balancing: Balancing the P3 Payload | Phase One.



> **Note**
>
> **The detailed step-by-step balance procedure is still being written** (assigned to @franco) and will be added in a future revision, along with a photo sequence and a final settings-table reference (tracked in the future-release backlog and `/IMAGE_TODO.md`). Until then, follow the Phase One instructional video linked above.

> **Tip**
>
> **Record and photograph the settings** (add to a settings table) so you have reference values for a stable gimbal.

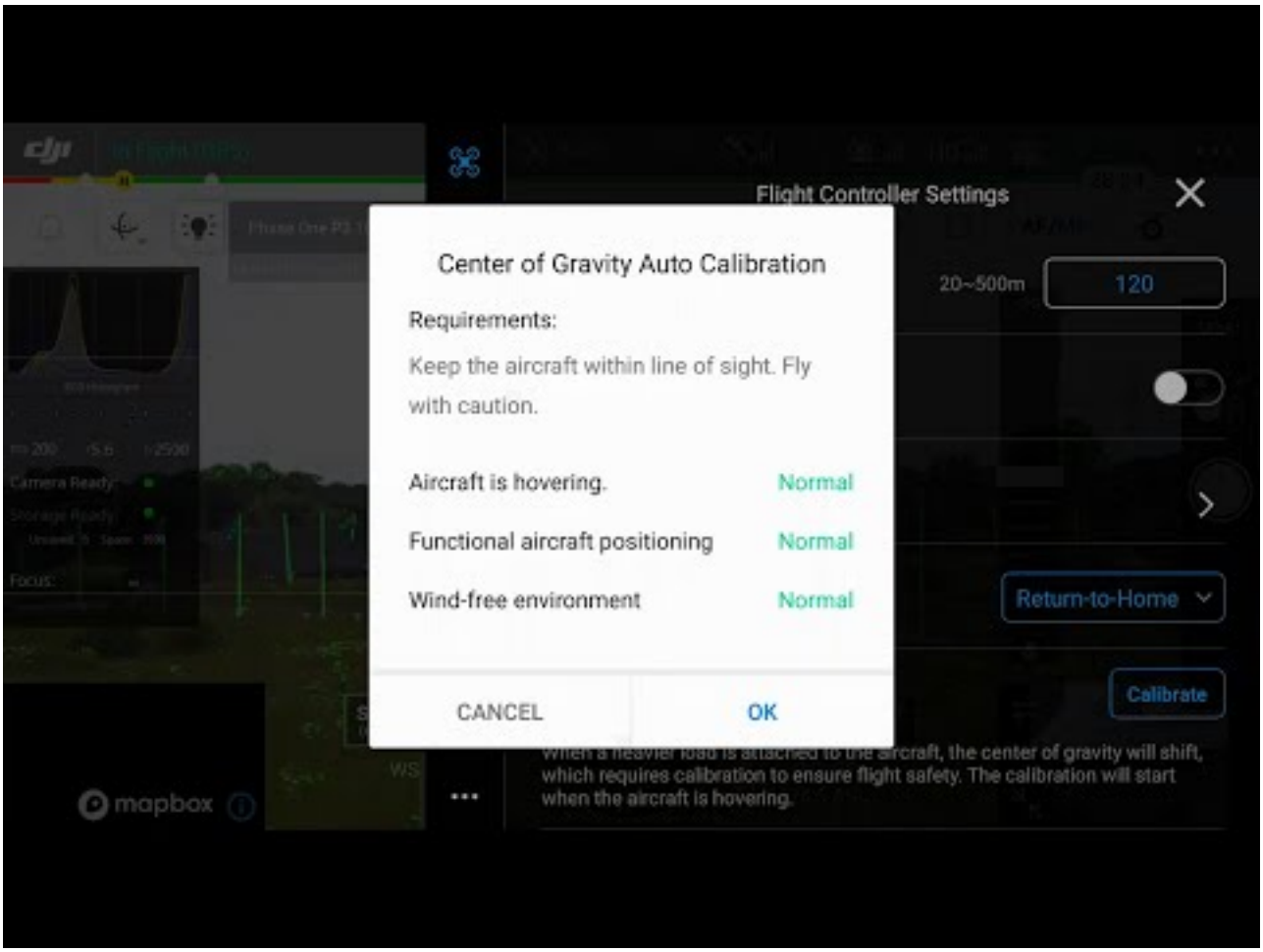
Drone Centre of Gravity Calibration

The centre of gravity will shift when the aircraft's payloads change. To ensure stable flight, it is required to recalibrate the aircraft's centre of gravity when a new payload is installed.

> **Warning**
>
> Calibrate in a windless environment. Make sure that the aircraft is hovering and there is a strong GNSS signal

during calibration. Maintain visual line of sight of the aircraft and pay attention to flight safety.

Refer to this video for more information: P3 Payload installation Part 8 | Phase One.



Note

Image needed (screenshot + photo): DJI Pilot 2 UI screenshot of the **Center of Gravity Auto Calibration** section (showing the Calibrate button and expected post-calibration prompt), plus a field photo of the aircraft in a windless environment in the hover state during calibration.

1. Go to **Flight Controller Settings** in the DJI Pilot 2 app.
2. Tap **Calibrate** in the **Center of Gravity Auto Calibration** section.
3. The Aircraft Status Indicators will glow solid purple during calibration.
4. There will be a prompt in the app after calibration is completed.

Set up custom camera in DJI Pilot 2

Section to be completed — document the steps to register the Phase One iXM-GS120 (80 mm) as a custom camera profile in DJI Pilot 2, including any required focal length, sensor size, and trigger configuration values.